Paper 2 Solution

1(a)	Area = 280 x 230 = 64400 mm^2 $\frac{\Delta A}{A} = \frac{\Delta L}{L} + \frac{\Delta W}{W}$ $\frac{\Delta A}{A} = \frac{1}{280} + \frac{2}{100}$ = 0.0236 ΔA = 0.0236 X 64400 = 2000 mm^2 A = 64000 ± 2000 mm^2	M1 M1 A1
1(b)	Accuracy refers to the closeness between the measured value 6.4 x 10^4 and the accepted or true value 8.00 x 10^4. ***As percentage difference of*** $\frac{80000-64400}{80000}$ ***x 100% = 19.5% is big,*** the measurement is not accurate***. *Will not penalize students if percentage difference is not calculated.*** Precision refers to the degree of scattering among repeated measurements. **As the maximum percentage uncertainty of** $\frac{2000}{64000}$ **x 100% =3.1% is small,** *the measurement is precise*. ****Will not penalize students if percentage difference is not calculated.***	B1 B1
2(a)(i)	Work is done when a force moves its point of application in the direction of the force.	B1